RSS-Based Access Point Localization Experiment

1 Experimental Setup

This experiment evaluates RSS-based localization for a single Wi-Fi access point inside the Faculty of Engineering at Ferdowsi University of Mashhad. The target access point was located in a corridor area of the building. An independently measured coordinate was kept only as ground truth for final error evaluation:

`36 deg 18’46.1"N, 59 deg 31’34.9"E`

which corresponds to:

`latitude = 36.3128055556,  longitude = 59.5263611111`.

The experiment uses the pivot RSS table as the only RSS data source:

`data/pivot_signal_table.csv`

The floor map was obtained from:

`data/map.json`

The room and corridor outlines were drawn from the `map_space_coordinate` table in the map JSON file.

2 Dataset Description

Each row in the pivot table contains the measurement position and the RSS values observed from multiple Wi-Fi access points. A value of $-100\,\mathrm{dBm}$ was treated as a missing or non-detected RSS observation. For the selected target access point, the number of valid RSS samples was 1069.

The selected access point was:

`BSSID = b28ba99a8d2a`

Table 1: Summary of RSS samples for the selected access point.

Quantity	Value
Valid RSS samples	1069
Maximum RSS	$-29.0\,\mathrm{dBm}$
Mean RSS	$-72.89\,\mathrm{dBm}$
RSS-weighted centroid error from ground truth	$1.37\,\mathrm{m}$

Figure 1 shows the RSS measurements of the selected access point overlaid on the building map. Stronger RSS values are shown in green, while weaker RSS values are shown in red.

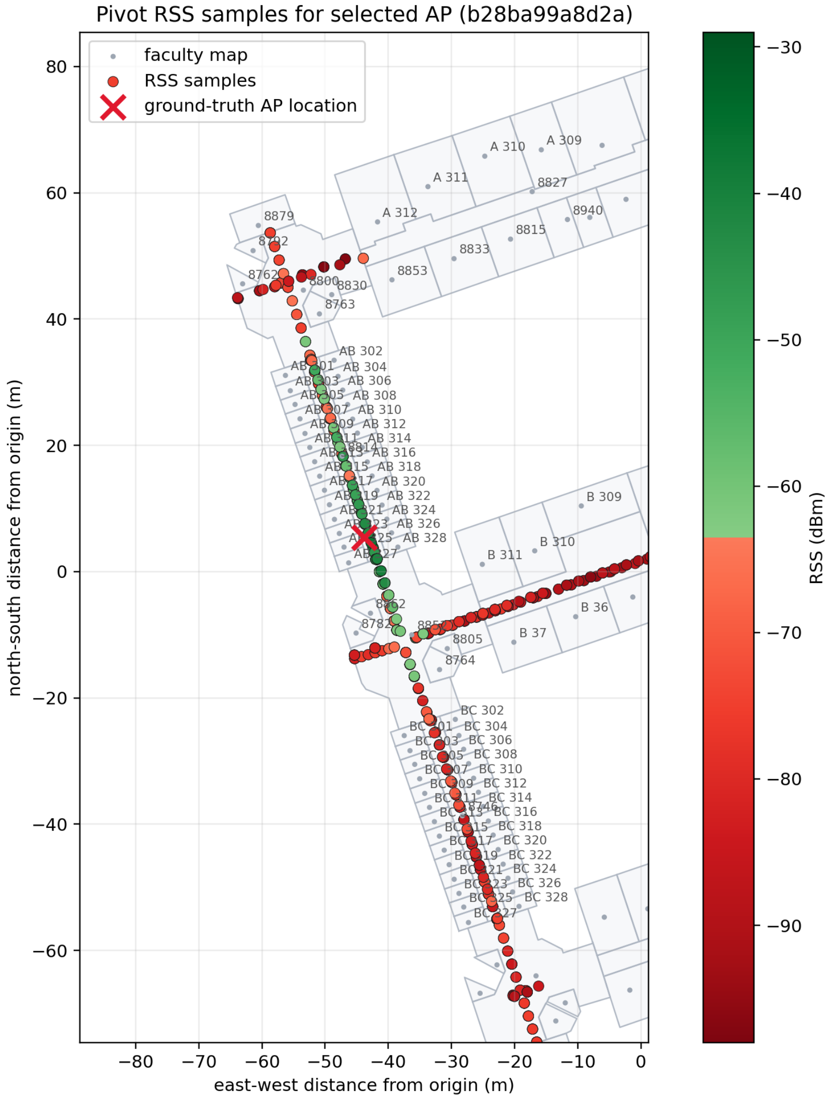

Figure 1: RSS samples of the selected access point overlaid on the faculty map.

3 Access Point Selection

To identify the access point corresponding to the target corridor location, all RSS columns in the pivot table were examined. For each access point, only valid RSS observations, i.e., samples with RSS greater than $-100\,\mathrm{dBm}$, were used. An RSS-weighted spatial centroid was then computed for each candidate access point.

For a sample with RSS value r_i, the weight was defined as:

$$w_i = 10^{(r_i - r_{\max})/10}, \tag{1}$$

where $r_{\max}$ is the strongest RSS observed for that access point. The weighted centroid was computed as:

$$x_c = \frac{\sum_i w_i x_i}{\sum_i w_i}, \qquad y_c = \frac{\sum_i w_i y_i}{\sum_i w_i}. \tag{2}$$

The selected access point was `b28ba99a8d2a`. The ground-truth coordinate was not used as an optimization input; it was used only to report the final localization error.

4 RSS Fitting Model

After selecting the target access point, its location was estimated by fitting a log-distance path-loss model to the RSS measurements. The model is:

$$RSS(d) = RSS_{1m} - 10n\log_{10}(d), \tag{3}$$

where RSS_{1m} is the estimated RSS at one meter, n is the path-loss exponent, and d is the Euclidean distance between a measurement location and the estimated access point position.

The fitted parameter vector was:

$$\theta = (x_{AP}, y_{AP}, RSS_{1m}, n). \tag{4}$$

The fitting procedure was performed step by step as follows:

1. Measurement coordinates were first converted from latitude and longitude to a local Cartesian coordinate system in meters.
2. The valid RSS samples of the selected AP were retained, and missing detections were excluded.
3. The initial AP position was set to the measurement point with the maximum RSS value. Therefore, the ground-truth coordinate was not used to initialize the optimization.
4. The initial RSS_{1m} value was set from a high RSS percentile, and the initial path-loss exponent was set to $n = 2$.
5. For each candidate AP position and model parameter set, distances from the candidate AP to all measurement points were computed, and RSS values were predicted using the log-distance model.
6. The optimizer minimized the residual between measured RSS and model-predicted RSS using nonlinear least squares with a robust `soft_l1` loss.
7. After convergence, the estimated AP position was converted back to latitude and longitude. The ground-truth coordinate was used only at this final stage to compute localization error.

5 Localization Results

Table 2 summarizes the localization and model-fitting results.

Table 2: Estimated AP location and fitted RSS model parameters.

Quantity	Value
Initial latitude	36.3127978000
Initial longitude	59.5263734000
Estimated latitude	36.3128106176
Estimated longitude	59.5263414105
Distance from ground truth	1.85 m
Estimated RSS_{1m}	−28.36 dBm
Path-loss exponent n	3.22
RSS RMSE	8.53 dB
RSS MAE	6.91 dB

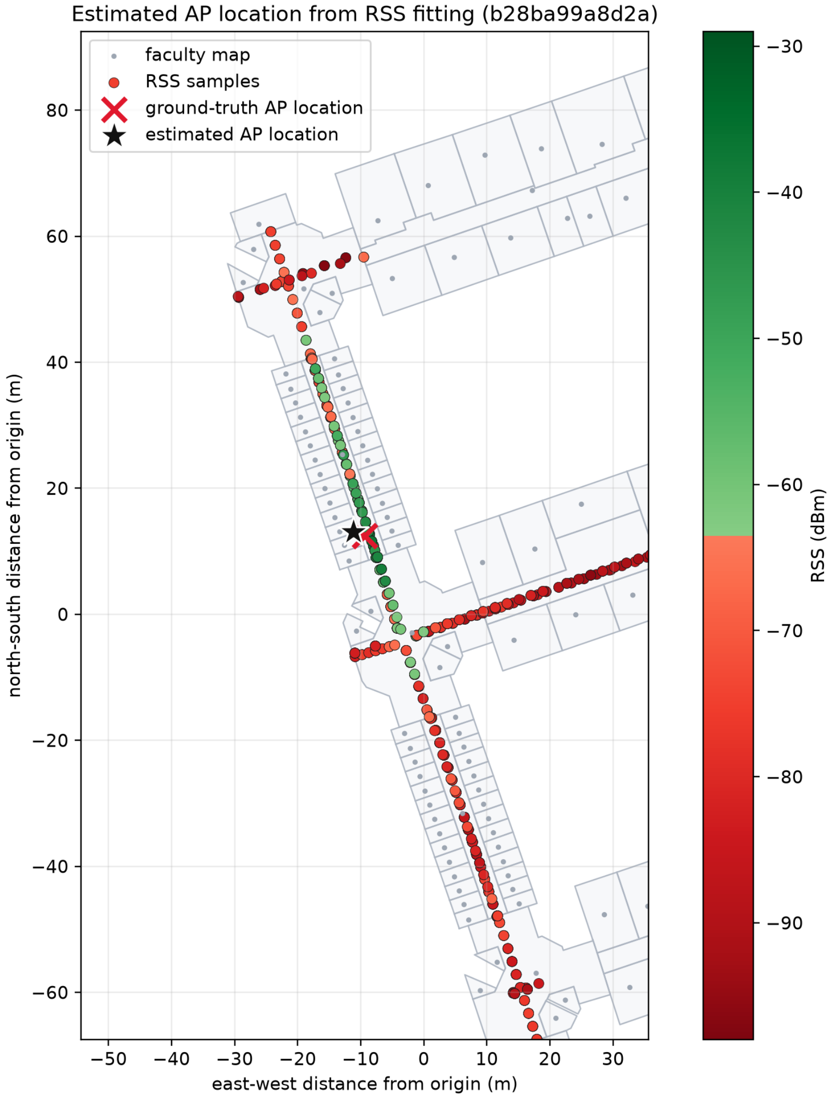

Figure 2: Reference and estimated access point locations on the faculty map.

Figure 2 shows the ground-truth AP coordinate and the estimated AP position on the building map. The red cross indicates the ground-truth coordinate, and the black star indicates the estimated AP location.

The estimated position was approximately 1.85 m away from the ground-truth coordinate. Considering the high variability of indoor RSS measurements, this result indicates that the fitted log-distance model can provide a practical approximation of the access point location when sufficient RSS samples are available around the target area.